



Operating Instructions

Diesel engine-generator set
mtu 10V1600 DS500 / DG10V1600A1I
mtu 10V1600 DS540 / DG10V1600A2I
mtu 12V1600 DS650 / DG12V1600A1I
mtu 12V1600 DS715 / DG12V1600A2I
Application groups 3B, 3D, 3E and 3F

Without engine-generator set controller

SS180184/00E

System designation	Engine model	Application group
mtu 10V1600 DS500 / DG10V1600A1I	10V1600G10F	3B, continuous service, variable load, ICXN
mtu 10V1600 DS540 / DG10V1600A2I	10V1600G20F	3B, continuous service, variable load, ICXN
mtu 12V1600 DS650 / DG12V1600A1I	12V1600G10F	3B, continuous service, variable load, ICXN
mtu 12V1600 DS715 / DG12V1600A2I	12V1600G20F	3B, continuous service, variable load, ICXN
mtu 10V1600 DS500 / DG10V1600A1I	10V1600G70F	3D, emergency standby power, IFN
mtu 10V1600 DS540 / DG10V1600A2I	10V1600G80F	3D, emergency standby power, IFN
mtu 12V1600 DS650 / DG12V1600A1I	12V1600G70F	3D, emergency standby power, IFN
mtu 12V1600 DS715 / DG12V1600A2I	12V1600G80F	3D, emergency standby power, IFN
mtu 10V1600 DS500 / DG10V1600A1I	10V1600G10F	3E, emergency service, ICXN
mtu 10V1600 DS540 / DG10V1600A2I	10V1600G20F	3E, emergency service, ICXN
mtu 12V1600 DS650 / DG12V1600A1I	12V1600G10F	3E, emergency service, ICXN
mtu 12V1600 DS715 / DG12V1600A2I	12V1600G20F	3E, emergency service, ICXN
mtu 10V1600 DS500 / DG10V1600A1I	10V1600G10F	3F, heavy duty for DCP, unrestricted, ICXN
mtu 10V1600 DS540 / DG10V1600A2I	10V1600G20F	3F, heavy duty for DCP, unrestricted, ICXN
mtu 12V1600 DS650 / DG12V1600A1I	12V1600G10F	3F, heavy duty for DCP, unrestricted, ICXN
mtu 12V1600 DS715 / DG12V1600A2I	12V1600G20F	3F, heavy duty for DCP, unrestricted, ICXN

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

© Copyright Rolls-Royce Solutions

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of Rolls-Royce Solutions. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage and/or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

Table of Contents

1	Preface		5.4.1 Configuration without engine-generator set controller (control version 1+)	58
1.1	Preface	6	5.4.2 Control cabinet	59
2	Safety		5.4.3 Smart Connect	61
2.1	Important provisions for all products	7	5.5 Fuel System	63
2.2	Intended use of engine-generator sets	8	5.5.1 Fuel system with common-rail injection	63
2.3	Personnel and organizational requirements	10	5.5.2 Fuel prefilter - Installation position	65
2.4	Safety regulations for commissioning and operation	11	5.5.3 Fuel prefilter	66
2.5	Safety regulations for assembly, maintenance, and repair work	13	5.5.4 Fuel prefilter with water separator	68
2.6	Fire and environmental protection, indirect materials	17	5.6 Charge-Air System and Exhaust System	70
2.7	Standards for warning messages in the text and highlighted information	20	5.6.1 Charge-air and exhaust system	70
2.8	Safety signs and markings	21	5.6.2 Air filter	72
2.9	Safety signs on engine-generator set - Overview	26	5.6.3 Heavy-duty air filter	74
2.10	Protective devices on engine-generator set - Overview	33	5.6.4 Exhaust silencer	77
3	Transport		5.7 Lube Oil System	78
3.1	Transport	34	5.7.1 Lube oil system	78
3.2	Tie-down instructions	36	5.7.2 Lube oil extraction pump	79
4	General Information		5.8 Cooling System	80
4.1	Data acquisition and data transmission	38	5.8.1 Cooling system	80
5	Functional Description		5.8.2 Preheater (2.5 kW)	82
5.1	Engine-Generator Set	39	5.9 Electrical System	84
5.1.1	Engine-generator set - Application groups	39	5.9.1 Starter battery	84
5.1.2	Engine-generator set - Nameplate and designation	41	5.9.2 Battery charger	86
5.1.3	Engine-generator set - Standard scope of delivery	44	5.9.3 Battery disconnecter	88
5.1.4	Engine-generator set - Optional scope of delivery	47	6 Technical Data	
5.2	Engine	50	6.1 Reference tables for technical data	89
5.2.1	Engine - Overview	50	6.2 Technical data mtu 10V1600 DS500, application group 3B	90
5.2.2	Engine side and cylinder designations	51	6.3 Technical data mtu 10V1600 DS540, application group 3B	92
5.2.3	Sensors - Overview	52	6.4 Technical data mtu 12V1600 DS650, application group 3B	94
5.2.4	Firing order	54	6.5 Technical data mtu 12V1600 DS715, application group 3B	96
5.3	Generator	55	6.6 Technical data mtu 10V1600 DS500, application group 3D	98
5.3.1	Generator - General information	55	6.7 Technical data mtu 10V1600 DS540, application group 3D	100
5.3.2	Voltage regulator D350	57	6.8 Technical data mtu 12V1600 DS650, application group 3D	102
5.4	Monitoring and Control	58	6.9 Technical data mtu 12V1600 DS715, application group 3D	104
			6.10 Technical data mtu 10V1600 DS500, application group 3E	106
			6.11 Technical data mtu 10V1600 DS540, application group 3E	108

6.12	Technical data mtu 12V1600 DS650, application group 3E	110	10.4	Engine	169
6.13	Technical data mtu 12V1600 DS715, application group 3E	112	10.4.1	Engine - Barring with starting system	169
6.14	Technical data mtu 10V1600 DS500, application group 3F	114	10.5	Generator	170
6.15	Technical data mtu 10V1600 DS540, application group 3F	116	10.5.1	Leroy-Somer Generator	170
6.16	Technical data mtu 12V1600 DS650, application group 3F	118	10.5.1.1	Generator - Winding temperature check	170
6.17	Technical data mtu 12V1600 DS715, application group 3F	120	10.5.1.2	Generator - Cleaning air inlet and air outlet	171
			10.5.1.3	Generator - Cleaning terminal box and checking screw connections for secure seating	172
7	Operation		10.5.1.4	Generator - Bearings securing screws check	173
7.1	Chapter overview	122	10.5.1.5	Generator - Check of protective equipment	175
7.2	Lockout/tagout procedures	123	10.5.1.6	Generator - Diode cleaning	176
7.3	Unlocking procedure	125	10.5.1.7	Generator - Re-tightening securing screws of diodes	177
7.4	Preparations for putting into operation after extended out-of-service periods	127	10.5.1.8	Generator - Drying windings	178
7.5	Engine-generator set - Start	130	10.5.1.9	Generator - Polarization index check	180
7.6	Engine-generator set - Shutdown	131	10.5.2	Generator mount - Measuring height of rubber element	181
7.7	Operational monitoring	132	10.5.3	Generator mounting - Securing screw firm seating check	182
7.8	Re-starting the engine following an automatic safety shutdown	133	10.5.4	Resilient mounts of generator mounting - Check	183
7.9	Emergency shutdown	134	10.6	Monitoring and Control	184
7.10	After shutdown - Taking the engine- generator set out of operation	135	10.6.1	Protective function (emergency stop button) - Check	184
8	Maintenance		10.7	Belt Drive	185
8.1	Noncyclical maintenance tasks	136	10.7.1	Drive belt - Condition check	185
8.2	Maintenance Schedule Task Reference Table [QL1]	137	10.7.2	Drive belt - Tension check	186
9	Troubleshooting		10.7.3	Drive belt - Replacement	188
9.1	Smart Connect fault messages	139	10.7.4	Drive belt - Adjustment	190
9.2	Smart Connect status displays	144	10.8	Valve Drive	191
9.3	ECU 8 fault messages	146	10.8.1	Valve clearance - Check and adjustment	191
10	Task Description		10.8.2	Cylinder head cover - Removal and installation	194
10.1	Chapter overview	161	10.9	Fuel System	197
10.2	Acoustic Enclosure	162	10.9.1	Fuel system - Venting	197
10.2.1	Acoustic enclosure - Check	162	10.10	Fuel Filter	198
10.2.2	Acoustic enclosure - External contamination check	163	10.10.1	Fuel filter - Replacement	198
10.2.3	Acoustic enclosure door - Checking seal	164	10.10.2	Fuel prefilter - Draining	199
10.2.4	Acoustic enclosure door - Checking hinges	165	10.10.3	Fuel prefilter with water separator - Draining	200
10.2.5	Grounding cable - Check	166	10.10.4	Fuel prefilter - Replacement	201
10.2.6	Grounding cable - Continuity test	167	10.10.5	Fuel prefilter with water separator - Replacement	202
10.3	Engine-Generator Set	168	10.11	Charge-Air Cooling	203
10.3.1	Engine-generator set - Test run	168	10.11.1	Intercooler - Checking condensate drain for coolant discharge and obstructions	203
			10.12	Air Filter	204
			10.12.1	Air filter - Replacement	204
			10.12.2	Air filter - Removal and installation	205
			10.12.3	Heavy-duty air filter - Removal and installation	206
			10.13	Air Intake	208

10.13.1	Service indicator - Signal ring position check	208	10.19.1	Preheater - Coolant line check	226
10.13.2	Heavy-duty air filter - Service indicator check	209	10.19.2	Preheater - Replacement	227
10.14	Exhaust System	210	10.19.3	Preheater - Thermostat replacement	229
10.14.1	Exhaust silencer - Condensate drain check	210	10.20	Wiring (General) for Engine/Gearbox/Unit	230
10.15	Starting Equipment	211	10.20.1	Engine cabling - Check	230
10.15.1	Starter - Condition check	211	10.21	Accessories for (Electronic) Engine Governor / Control System	231
10.16	Lube Oil System, Lube Oil Circuit	212	10.21.1	Engine governor and connectors - Cleaning	231
10.16.1	Engine oil - Level check	212	10.21.2	Engine governor - Checking plug connections	232
10.16.2	Engine oil - Change	213			
10.16.3	Engine oil filter - Replacement	215	11	Appendix A	
10.17	Coolant Circuit, General, High-Temperature Circuit	216	11.1	Abbreviations	233
10.17.1	Engine coolant - Level check	216	11.2	Conversion tables	236
10.17.2	Engine coolant - Change	217	11.3	Contact person/Service partner	240
10.17.3	Engine coolant - Draining	218			
10.17.4	Engine coolant - Filling	219	12	Appendix B	
10.17.5	Engine coolant pump - Relief bore check	221	12.1	Standard Tools and Special Tools	241
10.18	Cooling System	222	12.2	Consumables	245
10.18.1	Coolant cooler - External contamination and leak-tightness check	222	12.3	Index	246
10.18.2	Coolant cooler - Cleaning	223			
10.18.3	Fuel cooler - External contamination and leak-tightness check	225	13	Appendix C	
10.19	Preheater	226	13.1	Safety Data Sheet - Valve Regulated Lead Acid Batteries (VRLA)	251

1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

Store these Operating Instructions where they can be accessed by all persons entrusted with operating or servicing this product.

Before starting any work, read these Operating Instructions carefully. For safe operation of the product, always follow all safety instructions and work instructions.

Local accident prevention regulations and general safety regulations for the area of application of the product also apply.

These Operating Instructions form part of the product and refer to the latest design version of the product at the time of publication. They are subject to modification to reflect technical enhancements. Actual product may vary from figures shown.

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at: <http://www.mtu-solutions.com>

2.2 Intended use of engine-generator sets

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- As generating set, stationary applications, for on-site power generation (isolated operation), for power generation as mains backup system (standby power), as well as for mains parallel operation
- As generating set to produce electric power at the voltages / frequencies/ temperatures and the possible maximum power specified on the nameplate
- As generating set in closed rooms, which provide the required technical equipment, such as supply and exhaust air opening, exhaust pipe routing through the wall to the outside, and a fuel supply facility
- Within the admissible operating parameters in accordance with the (→ technical sales documents – product data – TVU data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and observance of all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques

The product shall not be operated in explosive atmospheres unless it fulfills the conditions for such use and an approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to use, ensure that the most recent version is available. The latest version can be found at:

- <http://www.mtu-solutions.com>

Reasonably foreseeable misuse

- The engine-generator set was not designed for outdoor usage.
- The engine-generator set was not designed for mobile applications.
- The engine-generator set must not be operated with fluids and lubricants which are not listed in the specifications.
- The engine-generator set was not designed for the usage as combined heat and power plant or in applications where special requirements apply, e.g. nuclear power plant, oil&gas, fracking.
- When the specified operating hours, i.e. the expected lifetime of the product, is exceeded, the need for maintenance is expected to increase.
- The engine-generator set shall not be operated at other voltages and frequencies than those specified on the nameplate.
- The total rated power consumption of all consumers connected to the engine-generator set shall not exceed the total output of the engine-generator set specified on the nameplate.

Without safety certification according to IBC or Eurocode8, it is not recommended to use engine-generator sets in regions with high seismic activity, with the influence coefficient being defined as 1.5.

If lower requirements apply, the seismic force can be estimated according to the FEMA 450 guide using the terms and definitions of DIN 4149 and of DIN EN 1998-1 (2010) respectively. If a_{gR} is lower than 0.39 m/s^2 , no proof of the seismic safety is required as per EN1998.

Modifications or conversions

Unauthorized changes to the product breach the conditions concerning its intended use and compromise safety.

Only modifications or conversions expressly authorized by the manufacturer are regarded as compliant in this sense. The manufacturer shall not be held liable for any damage resulting from unauthorized modifications or conversions.

2.3 Personnel and organizational requirements

Organizational measures of the user/manufactururer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.